Objective: Lab 6 is to Create a bare-metal "Hello World" application and run it on a debug configuration for a Cortex-A9 Fixed Virtual Platform (FVP) provided with ARM Development Studio.

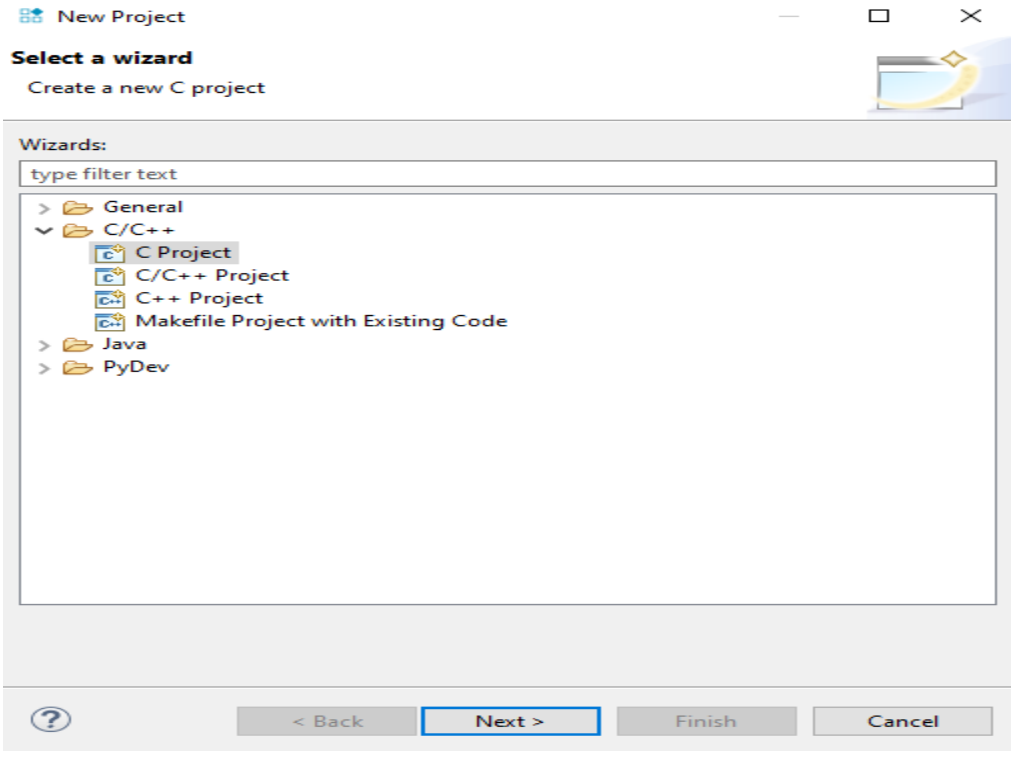
Document I followed to install the necessary things: [document to install arm ds and others eds tolls]

Procedure:

1. Install SoC EDS
2. Install ARM DS
3. Install Cygwin
4. Install MingGW
5. Install Linaro Bare Metal Toolchain
6. Installing tools on Windows.
7. Running the tools and activating the license
8. After activating the arm ds tool kit licensing ,Create your c project

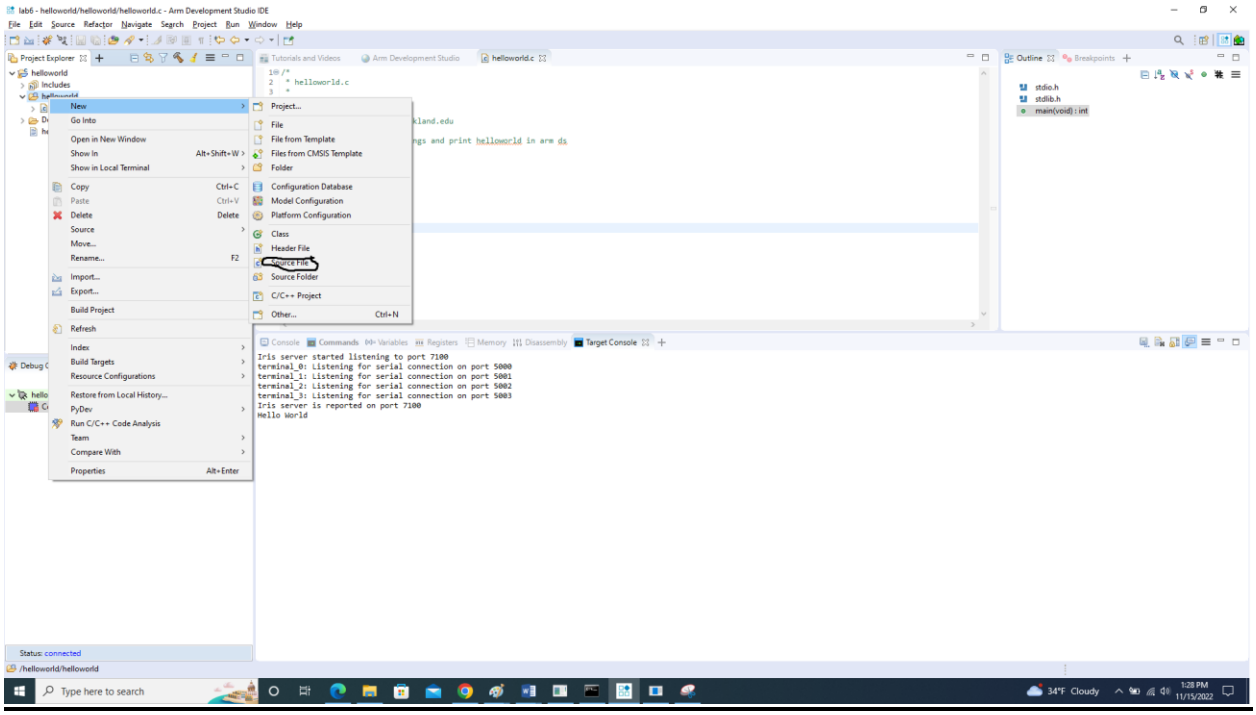
Steps to create a c project

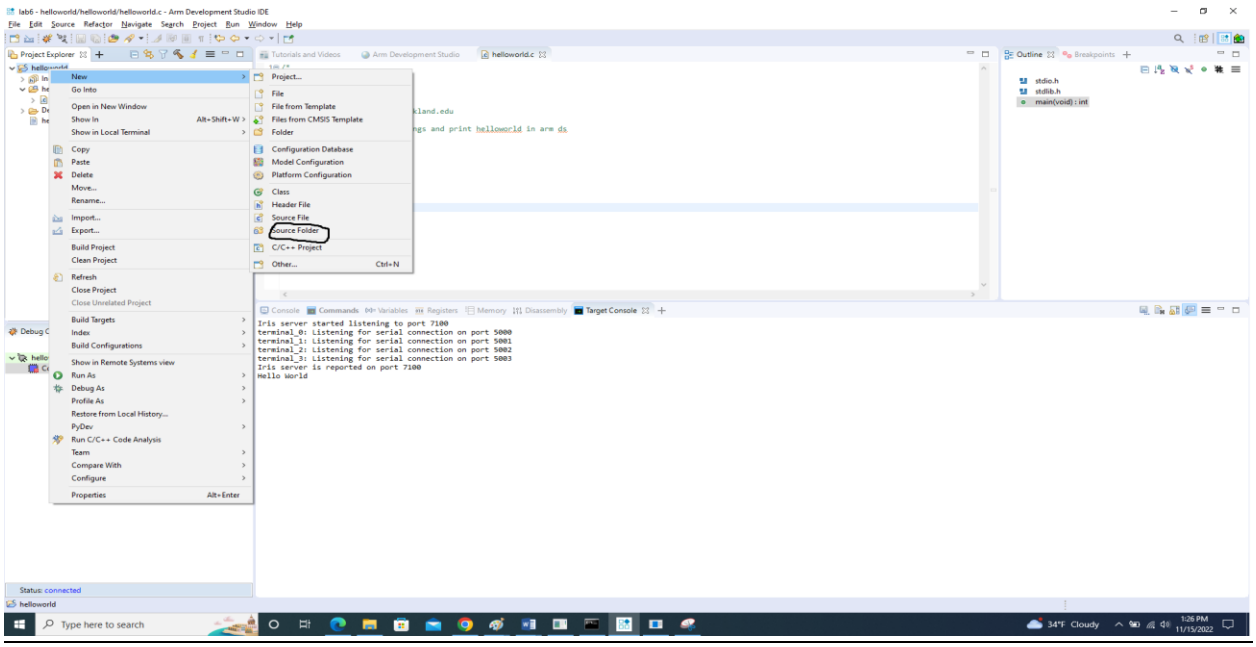
- Select File > New > Project to display the C Project dialog

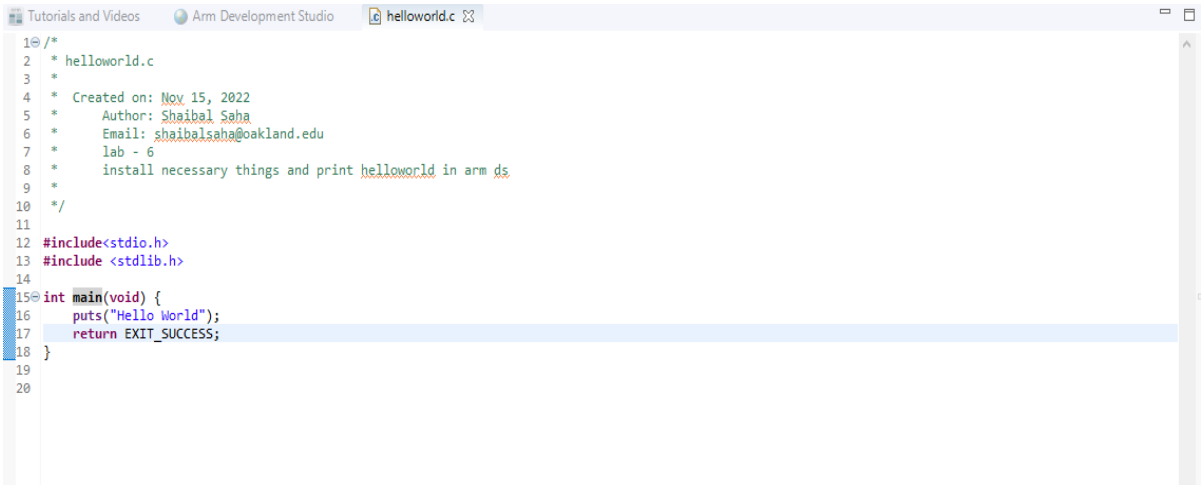


- In Project name field, enter the name of your project.
- Under Project type, select Empty Project.

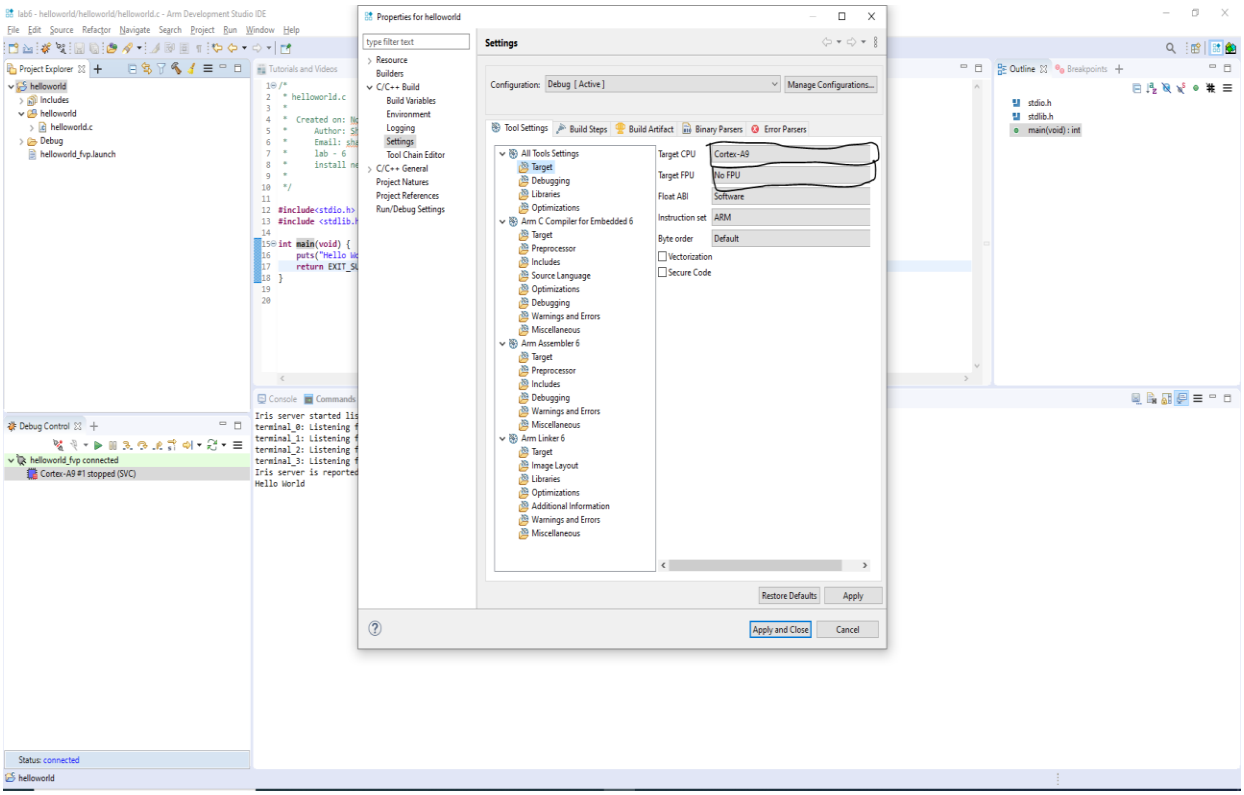
- Under Toolchains, select Arm Compiler 6.
- Click Finish to create a C project.
- You can view the project in the Project Explorer view
- Create the source folder and source file named .c and write c program which print “Hello world”

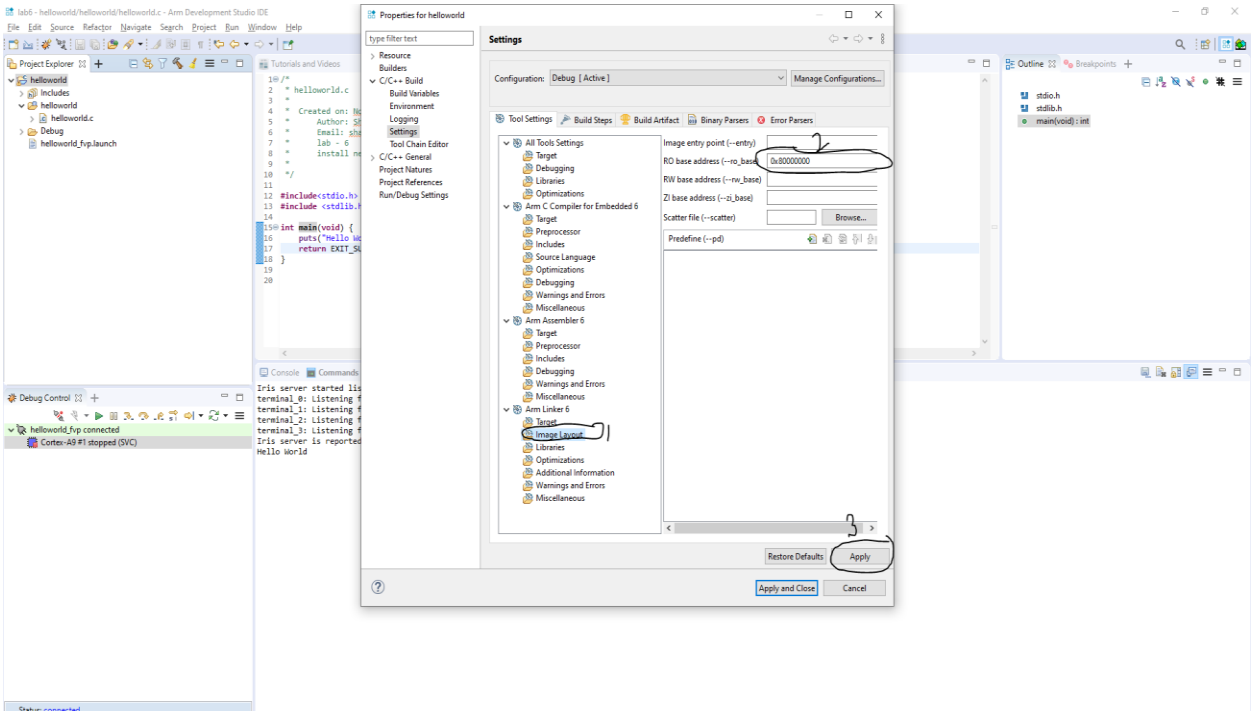




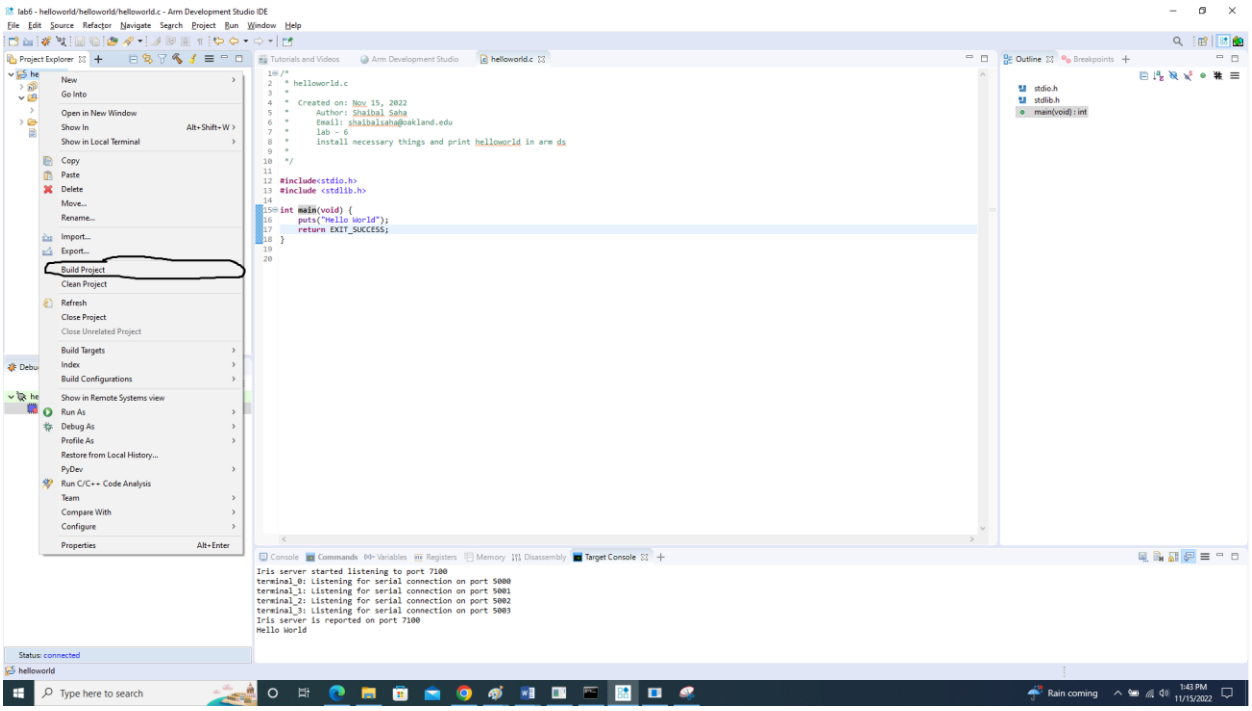


- After writing the project ->In Project Explorer, right-click on the project and select Properties
- **In the Properties dialog:**
  - ✓ Browse to C/C++ Build > Settings.
  - ✓ Under the Tool Settings tab, browse to All Tools Settings > Target, and from the Target CPU drop down select Cortex-A9.
  - ✓ Under the Tool Settings tab, browse to All Tools Settings > Target, and from the Target FPU drop down select No FPU.
  - ✓ Under the Tool Settings tab, browse to Arm Linker 6 > Image Layout.
  - ✓ In the RO base address (--ro_base) field, enter 0x80000000
  - ✓ Click apply and screenshots are below.

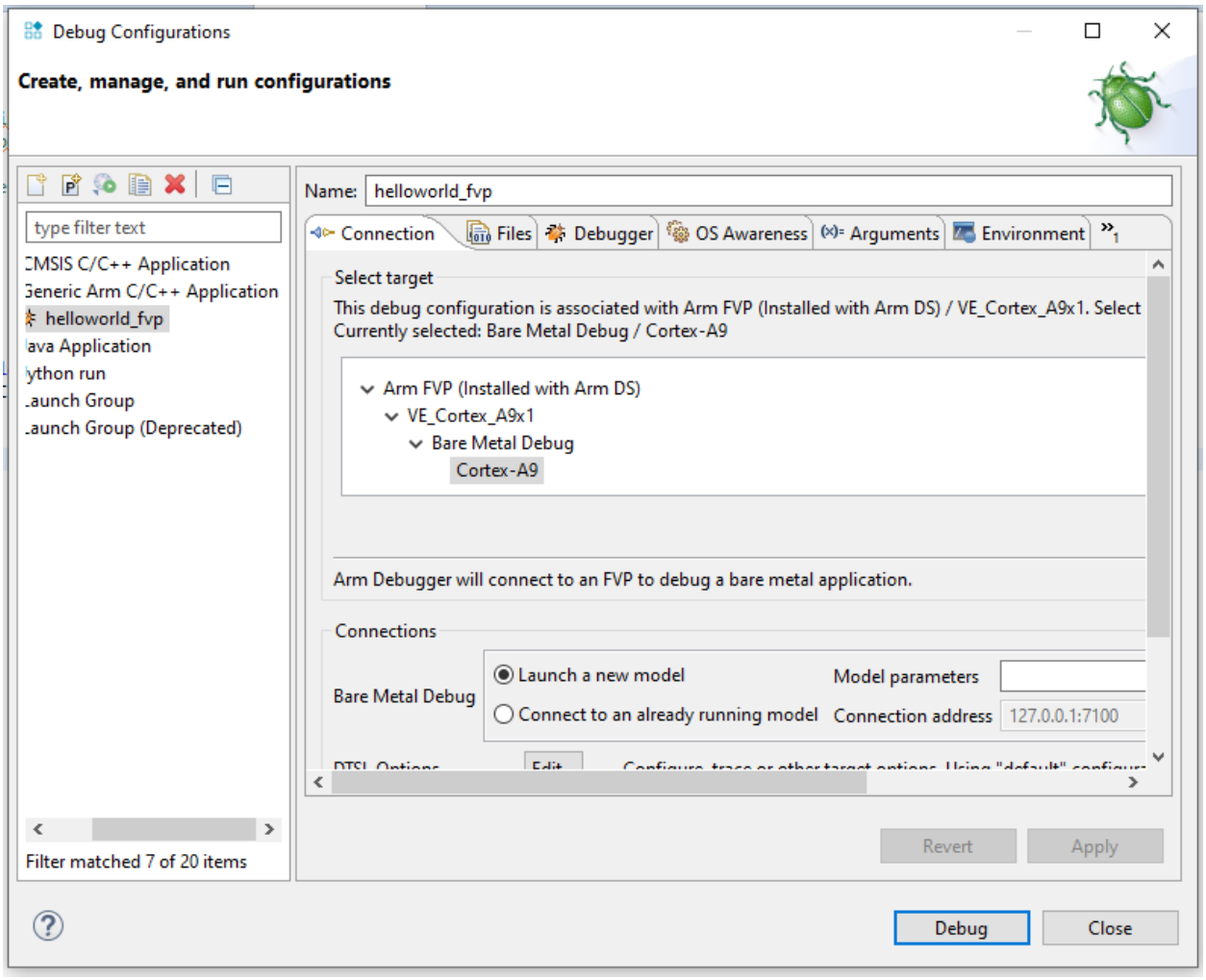


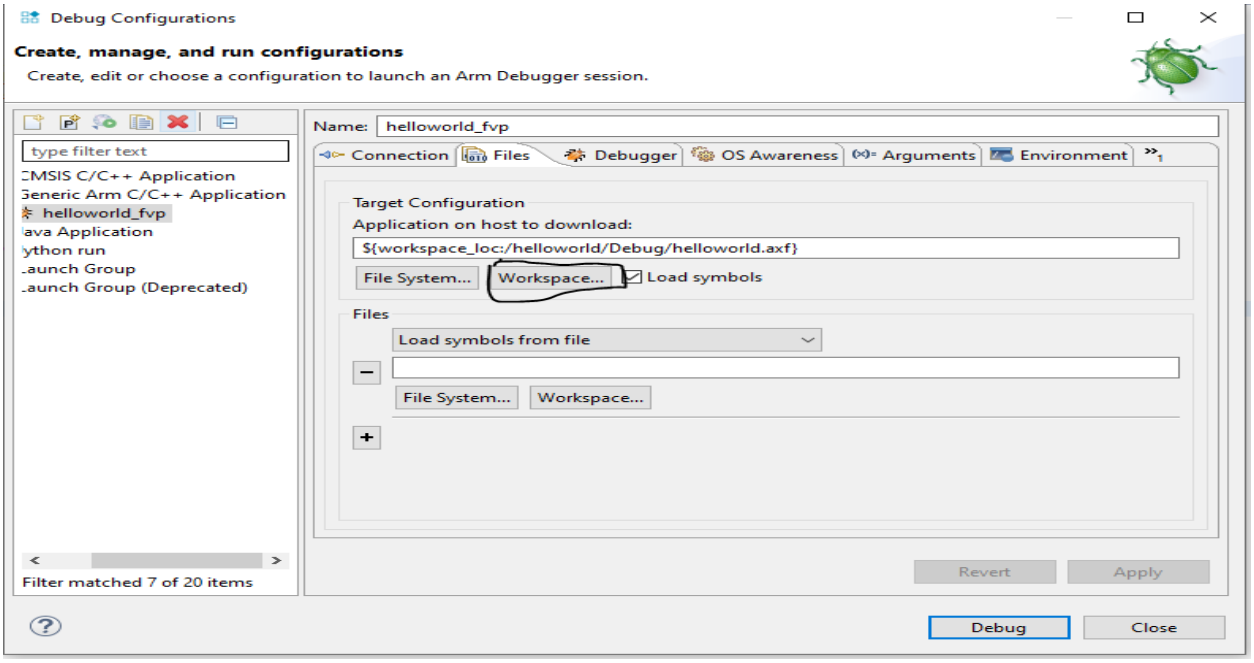


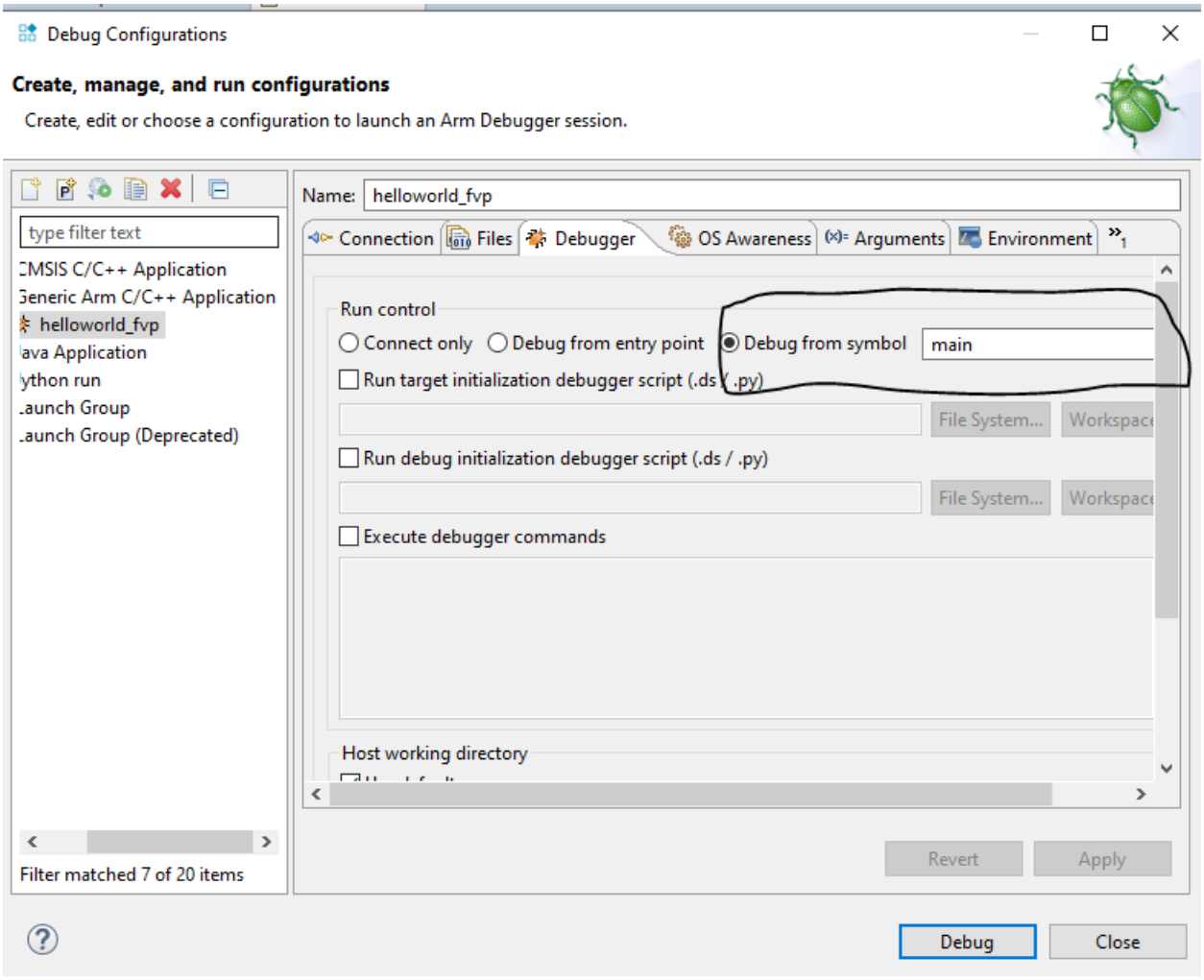
- Now build the project which will create your .axf file which I need later for FVP configuration.

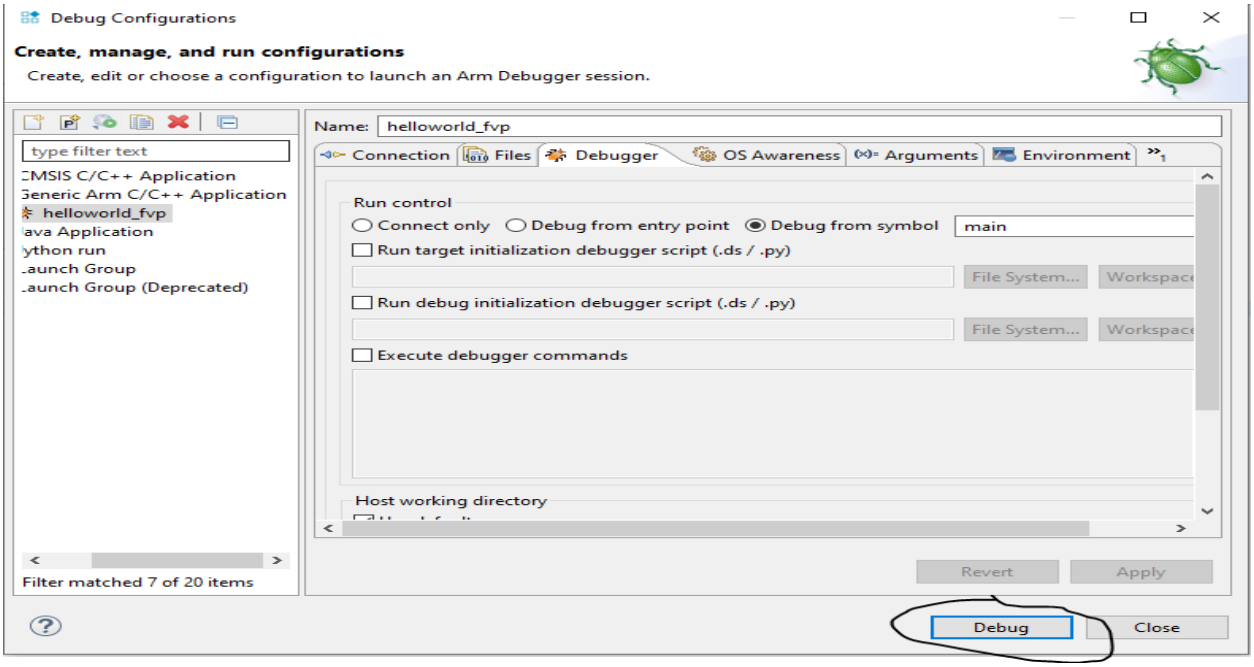


- For debug model connection create
  - ✓ Click the + sign in debug control window in the lower left side in the arm ds.
  - ✓ In the Connection tab, under Select Target, browse and select Arm FVP (Installed with Arm DS) > VE_Cortex_A9x1 -> Finish.
  - ✓ Select the Files tab, and under Target Configuration in the Application on host to download field, click Workspace.
  - ✓ The Workspace contains the **HelloWorld.axf** application file that created when built the Hello World project.
  - ✓ Select the Debugger tab, and ensure the Debug from symbol option is selected and set to **main**.

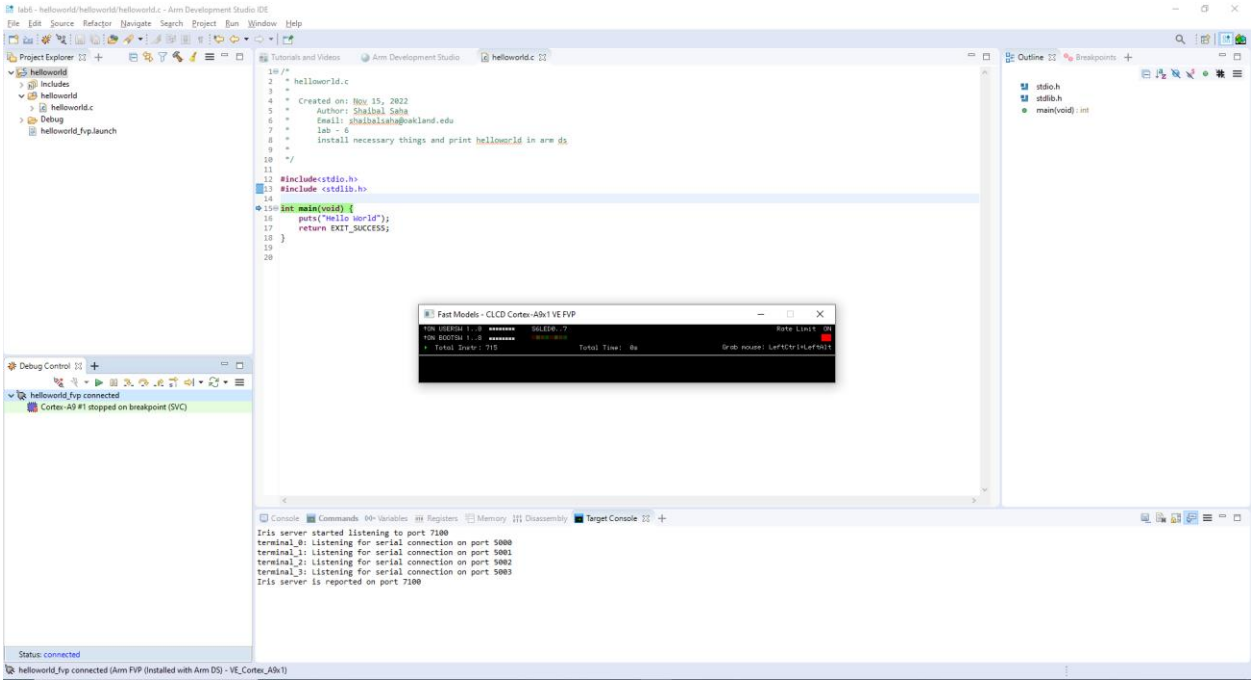

Debug Configurations
Create, manage, and run configurations
type filter text
CMSIS C/C++ Application
Generic Arm C/C++ Application
helloworld_fvp
Java Application
Python run
Launch Group
Launch Group (Deprecated)
Filter matched 7 of 20 items
Name: helloworld_fvp
Connection
Files
Debugger
OS Awareness
Arguments
Environment
Select target
This debug configuration is associated with Arm FVP (Installed with Arm DS) / VE_Cortex_A9x1. Select
Currently selected: Bare Metal Debug / Cortex-A9
Arm FVP (Installed with Arm DS)
VE_Cortex_A9x1
Bare Metal Debug
Cortex-A9
Arm Debugger will connect to an FVP to debug a bare metal application.
Connections
Bare Metal Debug
Launch a new model
Model parameters
Connect to an already running model
Connection address
127.0.0.1:7100
Edit
Revert
Apply
Debug
Close


Debug Configurations
Create, manage, and run configurations
Create, edit or choose a configuration to launch an Arm Debugger session.
type filter text
CMSIS C/C++ Application
Generic Arm C/C++ Application
helloworld_fvp
Java Application
Python run
Launch Group
Launch Group (Deprecated)
Filter matched 7 of 20 items
Name: helloworld_fvp
Connection
Files
Debugger
OS Awareness
Arguments
Environment
Target Configuration
Application on host to download:
${workspace_loc:/helloworld/Debug/helloworld.axf}
File System...
Workspace...
Load symbols
Files
Load symbols from file
File System...
Workspace...
Revert
Apply
Debug
Close

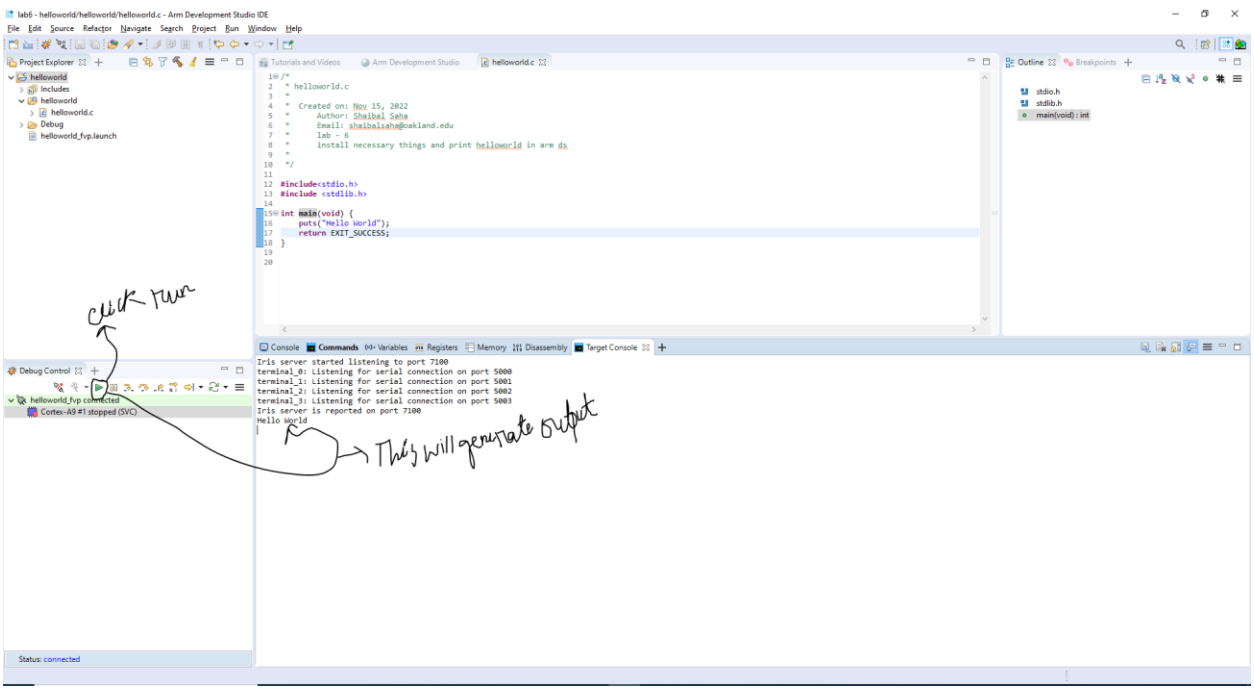

Debug Configurations
Create, manage, and run configurations
Create, edit or choose a configuration to launch an Arm Debugger session.
type filter text
CMSIS C/C++ Application
Generic Arm C/C++ Application
helloworld_fvp
ava Application
ython run
.aunch Group
.aunch Group (Deprecated)
Filter matched 7 of 20 items
Name: helloworld_fvp
Connection
Files
Debugger
OS Awareness
Arguments
Environment
Run control
Connect only
Debug from entry point
Debug from symbol
main
Run target initialization debugger script (.ds / .py)
File System...
Workspace
Run debug initialization debugger script (.ds / .py)
File System...
Workspace
Execute debugger commands
Host working directory
Revert
Apply
Debug
Close

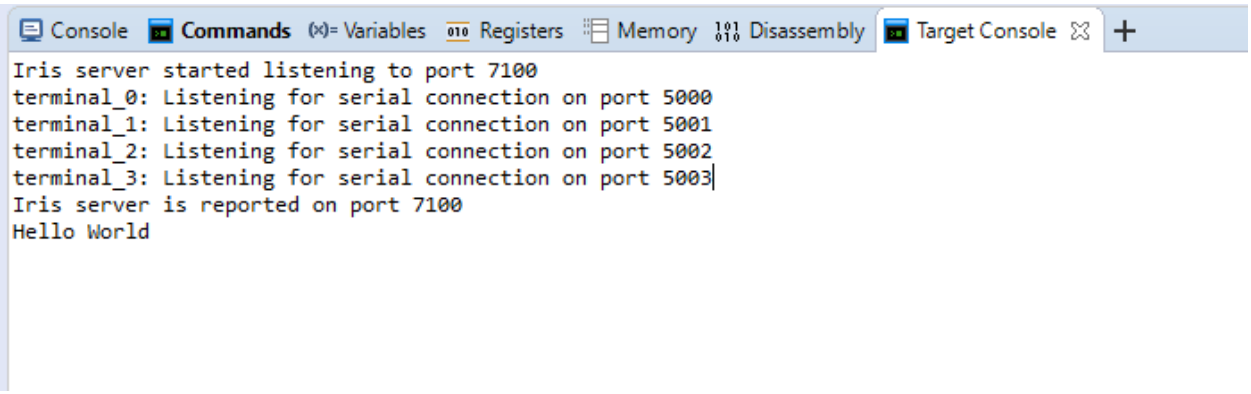

Debug Configurations
Create, manage, and run configurations
Create, edit or choose a configuration to launch an Arm Debugger session.
type filter text
CMSIS C/C++ Application
Generic Arm C/C++ Application
helloworld_fvp
ava Application
ython run
.aunch Group
.aunch Group (Deprecated)
Filter matched 7 of 20 items
Name: helloworld_fvp
Connection
Files
Debugger
OS Awareness
Arguments
Environment
Run control
Connect only
Debug from entry point
Debug from symbol
main
Run target initialization debugger script (.ds / .py)
File System...
Workspace
Run debug initialization debugger script (.ds / .py)
File System...
Workspace
Execute debugger commands
Host working directory
Revert
Apply
Debug
Close

- After click debug, it will connect to the model and displays the connection status in the Debug Control.



- Run the FVP





- So, it will show the output in the target console. We can also see the disassembly view that the loaded program in memory as assembler instructions at addresses.

NOTE: Source files are inside the lab6/helloworld folder